3 Connecting AI models with the Vercel AI SDK

This chapter covers

- An overview of the Vercel AI SDK for React/Next.js
- Incorporating streaming AI responses into React apps
- Separating AI and UI state in React/Next.js
- Enhancing conversational UIs with multimedia content

While building our minimal AI application, we encountered several challenges. One was *vendor lock-in*. Since our direct API calls created a tight coupling with a single provider, it would be difficult to switch services if we ever needed to. A second challenge concerned streaming responses: delivering AI responses in real time is a critical user experience feature that requires significant, nontrivial architectural changes. Finally, there is the problem of *state management*. As applications grow, managing both UI state and the conversational state (memory) becomes a complex task requiring a robust, scalable solution.

Rather than building custom solutions for each of these problems, a better approach is to use a dedicated framework that provides these capabilities out of the